FIITJEE

JEE (Main)-2018

ALL INDIA TEST SERIES

PART TEST – II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

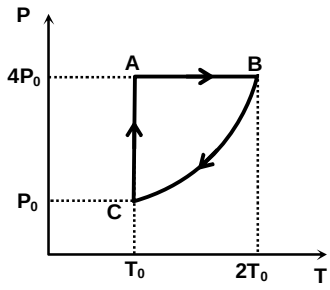
Physics

PART – I

SECTION – A
(One Options Correct Type)

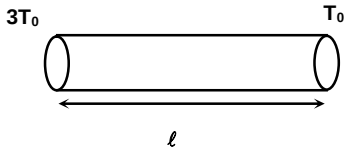
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Two moles of an ideal monoatomic gas is taken through a cyclic process as shown in the P-T diagram. In the process BC, PT^{-2} = constant. Then the ratio of heat absorbed and heat released by the gas during the process AB and process BC respectively is



(A) 2 (B) 3
(C) 5 (D) 6

2. Two ends of a rod of uniform cross sectional area are kept at temperatures $3T_0$ and T_0 as shown. Thermal conductivity of rod varies as $k = \alpha T$, (where α is a constant and T is absolute temperature). In steady state, the temperature of the middle section of the rod is



(A) $\sqrt{7}T_0$ (B) $\sqrt{5}T_0$
(C) $2T_0$ (D) $\sqrt{3}T_0$

3. An ideal gas is taken through a process $PT^3 =$ constant. The coefficient of thermal expansion of the gas in the given process is
(A) 1/T (B) 2/T
(C) 3/T (D) 4/T

4. 4 kg of ice at $-20°C$ is mixed with 5 kg of water at 40°C. The water content in the equilibrium mixture is (S_{water} = 1kcal/kg-C, S_{ice} = 0.5 kcal/kg-c, $L_{f(water)}$ = 80 kcal/kg)
(A) 6 kg (B) 7 kg
(C) 8 kg (D) 9 kg

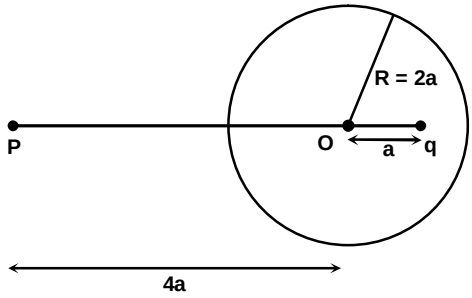
Space for Rough work

5. The molar heat capacity of an ideal gas in a process varies as $C = C_V + \alpha T^2$(where C_V is molar heat capacity at constant volume and α is a constant). Then the equation of the process is

(A) $Ve^{-\left(\frac{\alpha T^2}{2R}\right)} = \text{constant}$ (B) $Ve^{-\left(\frac{\alpha T^2}{R}\right)} = \text{constant}$

(C) $Ve^{-\left(\frac{2\alpha T^2}{R}\right)} = \text{constant}$ (D) $Ve^{-\left(\frac{3\alpha T^2}{2R}\right)} = \text{constant}$

6. A point charge 'q' is placed at distance 'a' from the centre of an uncharged thin spherical conducting shell of radius R = 2a. A point 'P' is located at a distance '4a' from the centre of the conducting shell as shown. The electric potential due to induced charge on the inner surface of the conducting shell at point 'P' is



(A) $\frac{kq}{5a}$ (B) $-\frac{kq}{5a}$

(C) $\frac{kq}{4a}$ (D) $-\frac{kq}{4a}$

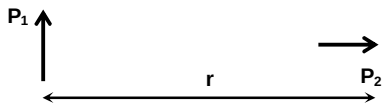
7. A solid sphere of radius 'R' density 'ρ', and specific heat 'S' initially at temperature T_0 Kelvin is suspended in a surrounding at temperature 0K. Then the time required to decrease the temperature of the sphere from T_0 to $\frac{T_0}{2}$ Kelvin is (Assume sphere behaves like a black body)

(A) $\frac{\rho SR}{9\sigma T_0^3}$ (B) $\frac{5\rho SR}{9\sigma T_0^3}$

(C) $\frac{7\rho SR}{9\sigma T_0^3}$ (D) $\frac{8\rho SR}{9\sigma T_0^3}$

Space for Rough work

8. Two point electric dipoles with dipole moments 'P_1' and 'P_2' are separated by a distance 'r' with their dipole axes mutually perpendicular as shown. The force of interaction between the dipoles is $\left(\text{where, } k = \frac{1}{4\pi\varepsilon_0}\right)$



(A) $\frac{2kP_1P_2}{r^4}$ (B) $\frac{3kP_1P_2}{r^4}$

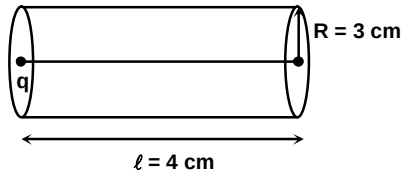
(C) $\frac{4kP_1P_2}{r^4}$ (D) $\frac{6kP_1P_2}{r^4}$

9. 4 moles of an ideal monoatomic gas is heated isobarically so that its absolute temperature increases 2 times. Then the entropy increment of the gas in this process is

(A) 285. J/k (B) 42.5 J/k

(C) 57.5 J/k (D) 76.5 J/k

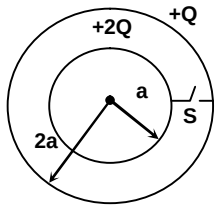
10. A point charge 'q' is placed at the centre of left circular end of a cylinder of length ℓ = 4 cm and radius R = 3 cm as shown. Then the electric flux through the curved surface of the cylinder is



(A) $\frac{q}{5\varepsilon_0}$ (B) $\frac{2q}{5\varepsilon_0}$

(C) $\frac{3q}{5\varepsilon_0}$ (D) $\frac{4q}{5\varepsilon_0}$

11. Two concentric spherical conducting shells of radii 'a' and '2a' are initially given charges +2Q and +Q respectively as shown. The total heat dissipated after switch 'S' has been closed is (where $k = \frac{1}{4\pi\varepsilon_0}$)

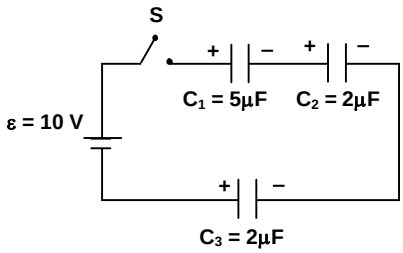


(A) $\frac{5kQ^2}{4a}$ (B) $\frac{kQ^2}{a}$

(C) $\frac{kQ^2}{2a}$ (D) $\frac{kQ^2}{4a}$

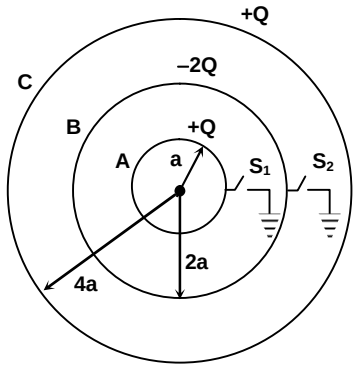
Space for Rough work

12. Three capacitors of capacitances 5 μF, 2μF and 2 μF are charged to 20 V, 30 V and 10 V respectively and then connected in the circuit with polarities as shown. The magnitude of charge flown through the battery after closing the switch 'S' is



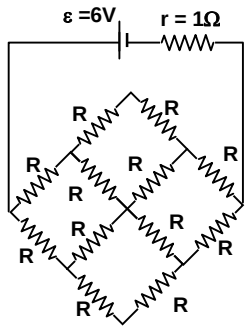
(A) 25 μC (B) 40 μC
(C) 50 μC (D) 75 μC

13. Three concentric spherical conducting shells A, B and C of radii a, 2a and 4a are initially given charges +Q, –2Q and +Q respectively. The charge on the middle spherical shell 'B' after switches S_1 and S_2 are simultaneously closed, will be



(A) –Q (B) $-\frac{Q}{2}$
(C) +Q (D) $\frac{+3Q}{2}$

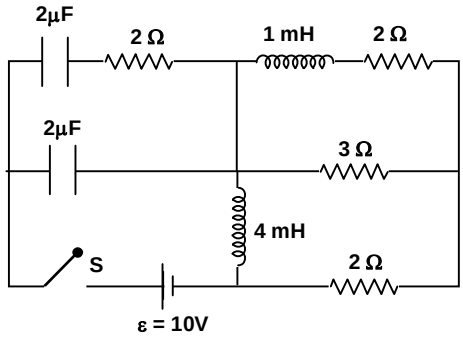
14. In the circuit shown the cell will deliver maximum power to the network if the value of 'R' is



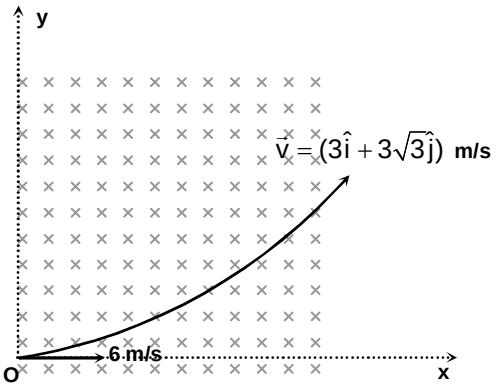
(A) $\frac{2}{3}\Omega$ (B) $\frac{4}{3}\Omega$
(C) $\frac{1}{3}\Omega$ (D) $\frac{5}{3}\Omega$

Space for Rough work

15. In the circuit shown, all the capacitors are initially uncharged. The current through the battery just after closing the switch 'S' is
(A) 2 A
(B) 4 A
(C) 5 A
(D) 10 A

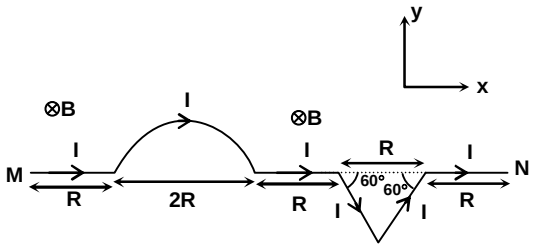


16. A particle of charge 'q' and mass 'm' enters a uniform magnetic field region $\vec{B} = -B\hat{k}$ at origin 'O' with a velocity $\vec{v}_0 = 6\hat{i}$ m/s and after some time it exits the magnetic field region with a velocity $\vec{v} = (3\hat{i} + 3\sqrt{3}\hat{j})$ m/s as shown. The time interval for which the particle has moved in the magnetic field region is



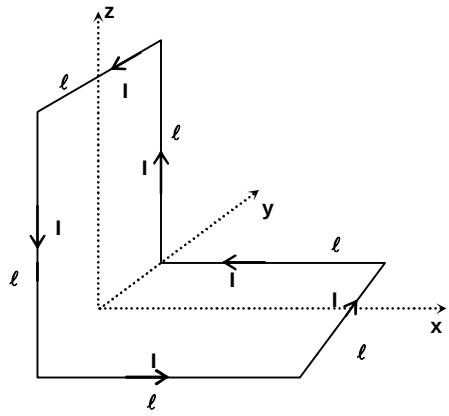
(A) $\frac{\pi m}{qB}$
(B) $\frac{\pi m}{2qB}$
(C) $\frac{\pi m}{3qB}$
(D) $\frac{\pi m}{4qB}$

17. A conducting wire MN carrying a current I is bent into the shape as shown and placed in xy plane. A uniform magnetic field $\vec{B} = -B\hat{k}$ is existing in the region. The net magnetic force experienced by the conducting wire MN is
(A) 2BIR
(B) 3BIR
(C) 4BIR
(D) 6BIR



Space for Rough work

18. A conducting loop carrying current 'I' is bent into two halves and placed in mutually perpendicular planes xy and yz planes as shown. A uniform magnetic field $\vec{B} = B\hat{j}$ is existing in the region. The net magnetic torque experienced by the loop is



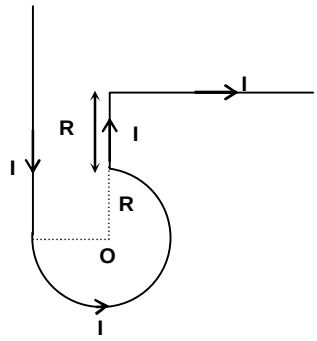
(A) $BI\ell^2(\hat{k} - \hat{i})$

(B) $BI\ell^2(\hat{i} + \hat{j})$

(C) $BI\ell^2(\hat{j} - \hat{k})$

(D) $BI\ell^2(\hat{i} + \hat{k})$

19. A conducting wire carrying a current I is bent into the shape as shown. The net magnetic field at the centre 'O' of the circular arc of radius 'R' is



(A) $\frac{\mu_0 I}{2R}\left(\frac{1+3\pi}{\pi}\right)$

(B) $\frac{\mu_0 I}{4R}\left(\frac{1+3\pi}{\pi}\right)$

(C) $\frac{\mu_0 I}{8R}\left(\frac{1+3\pi}{\pi}\right)$

(D) $\frac{\mu_0 I}{8R}\left(\frac{2+3\pi}{\pi}\right)$

20. An infinitely long cylindrical wire of radius 'R' is carrying a current with current density $j = \alpha r^3$ (where α is constant and 'r' is the radial distance from the axis of the wire). If the magnetic field at $r = \frac{R}{2}$ is 'B_1' and at r = 2R is 'B_2' then the ratio $\frac{B_2}{B_1}$ is

(A) 2

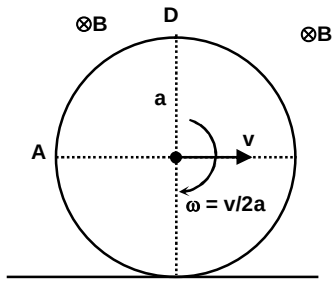
(B) 4

(C) 6

(D) 8

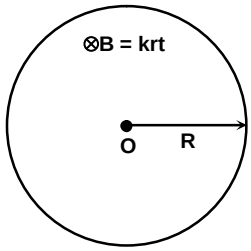
Space for Rough work

21. A circular conducting ring of radius 'a' is rolling with slipping on a horizontal surface as shown. A uniform magnetic field 'B' is existing perpendicular to the plane of motion of the ring. The emf induced between the points 'A' and 'D' of the ring is



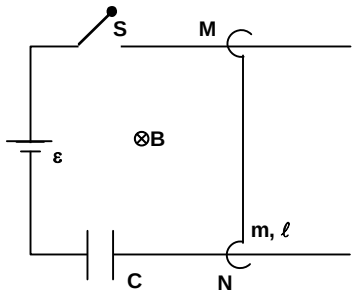
(A) $B\omega a^2$ (B) $2\,B\omega a^2$
(C) $4\,B\omega a^2$ (D) $6\,B\omega a^2$

22. A time varying magnetic field B = krt (where k is a constant, r is the radial distance from centre 'O') is existing in a circular region of radius 'R' as shown. If induced electric field at $r = \frac{R}{2}$ is E_1 and at r = 2R is E_2 then the ratio $\frac{E_2}{E_1}$ is



(A) 6 (B) 4
(C) 2 (D) 1

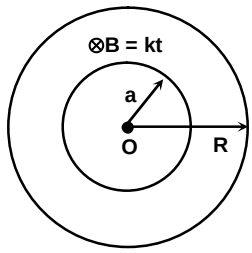
23. A conducting rod MN of mass 'm and length 'ℓ' is placed on parallel smooth conducting rails connected to an uncharged capacitor of capacitance 'C' and a battery of emf ε as shown. A uniform magnetic field 'B' is existing perpendicular to the plane of the rails. The steady state velocity acquired by the conducting rod MN after closing switch S is (neglect the resistance of the parallel rails and the conducting rod)



(A) $\frac{2CB\ell\varepsilon}{(m+CB^2\ell^2)}$ (B) $\frac{CB\ell\varepsilon}{(m+CB^2\ell^2)}$
(C) $\frac{CB\ell\varepsilon}{2(m+CB^2\ell^2)}$ (D) $\frac{CB\ell\varepsilon}{4(m+CB^2\ell^2)}$

Space for Rough work

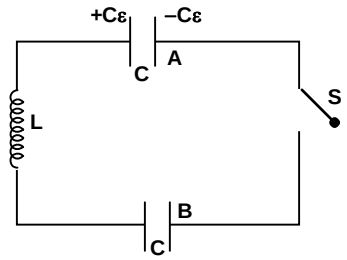
24. Charge 'q' is uniformly distributed along the length of a non-conducting circular ring of mass 'm' and radius 'a'. The ring is placed concentrically on a rough horizontal circular surface of radius 'R'. A time varying magnetic field B = kt (where k is constant) is existing perpendicular to the plane of circular region of radius 'R' as shown. The minimum coefficient of friction between the ring and the surface required to keep the ring stationary is



(A) $\frac{2kqa}{mg}$ (B) $\frac{kqa}{mg}$

(C) $\frac{kqa}{2mg}$ (D) $\frac{kqa}{4mg}$

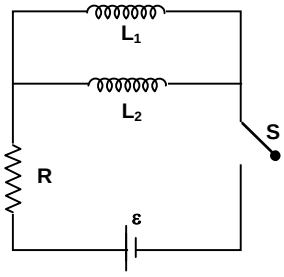
25. Initially capacitor 'A' is charged to a potential drop 'ε' and capacitor B is uncharged. At t = 0, switch 'S' is closed, then the maximum current through the inductor is



(A) $\varepsilon\sqrt{\frac{C}{2L}}$ (B) $\varepsilon\sqrt{\frac{C}{L}}$

(C) $2\varepsilon\sqrt{\frac{C}{L}}$ (D) $\frac{\varepsilon}{2}\sqrt{\frac{C}{L}}$

26. In the circuit shown, switch's is closed at t = 0. Then the ratio $\frac{U_1}{U_2}$ of potential energy stored in the inductors L_1 and L_2 in steady state is



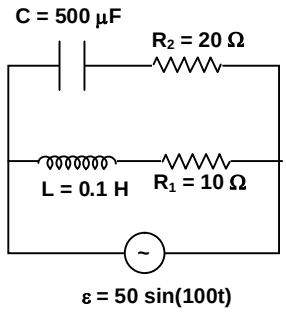
(A) $\frac{L_1}{L_2}$ (B) $\frac{L_2}{L_1}$

(C) $\left(\frac{L_1}{L_2}\right)^2$ (D) $\left(\frac{L_2}{L_1}\right)^2$

Space for Rough work

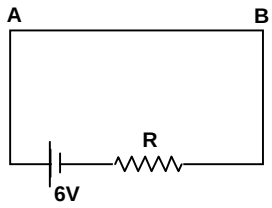
27. In the circuit shown the average power developed in the resistor 'R_1' is

(A) 31.25 W (B) 62.50 W
(C) 125 W (D) 250 W



28. A uniform potentiometer wire AB of length 100 cm has a resistance of 5 Ω and it is connected in series with an external resistance 'R' and a cell of emf 6V and negligible internal resistance. If a source of potential drop 2V is balanced against a length of 60 cm of the potentiometer wire, the value of resistance 'R' is

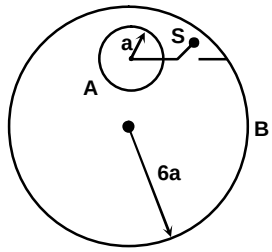
(A) 1Ω (B) 2Ω
(C) 4Ω (D) 6Ω



29. A small spherical conductor 'A' of radius 'a' is initially charged to a potential 'V' and then placed inside an uncharged spherical conducting shell 'B' of radius '6a' as shown. The potential of the spherical conductor 'A' after closing the switch 'S' is

(A) V (B) $\frac{V}{3}$

(C) $\frac{V}{2}$ (D) $\frac{V}{6}$



30. Two moles of an ideal diatomic gas is taken through a process VT^2 = constant so that its temperature increases by ΔT = 300 K. The ratio $\left(\frac{\Delta U}{\Delta Q}\right)$ of increase in internal energy and heat supplied to the gas during the process is

(A) 2 (B) 3
(C) 4 (D) 5

Space for Rough work

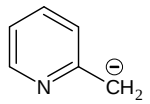
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

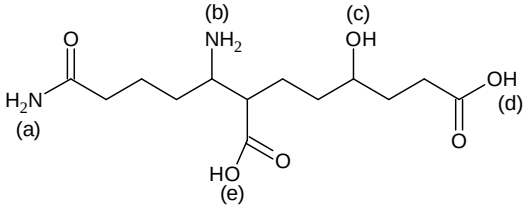
31. The number of resonating structures possible for



are?

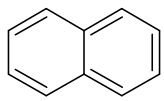
(A) 4
(B) 5
(C) 3
(D) 6

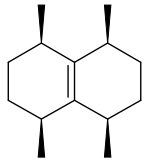
32. The most acidic portion among the following is:

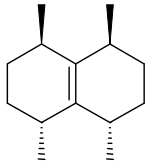


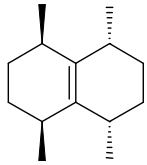
(A) a
(B) c
(C) e
(D) b

33. Which one of the following does not contain centre of symmetry?

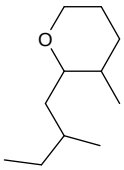
(A) 

(B) 

(C) 

(D) 

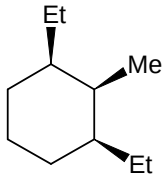
34. The total number of optical isomers possible for this compound are?

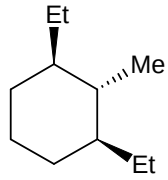


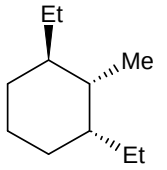
(A) 9
(B) 8
(C) 10
(D) 16

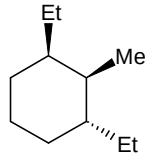
Space for Rough work

35. Which of the following has the least negative heat of combustion?

(A) 

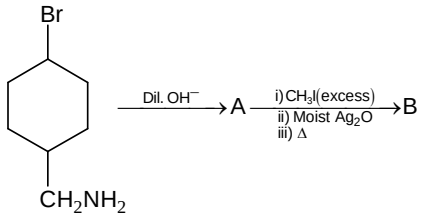
(B) 

(C) 

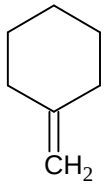
(D) 

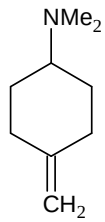
36. Which of the following agents is responsible for generating chlorine radicals into stratosphere?
(A) Smog (B) NO_2
(C) UV radiation (D) CFC

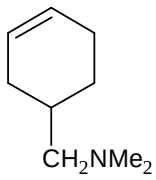
37. Which one is a copolymer?
(A) PVC (B) Polypropene
(C) Polystyrene (D) Glyptal

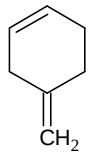
38. 

Identify the product in the following sequence of reaction?

(A) 

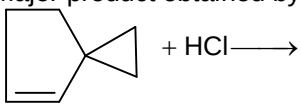
(B) 

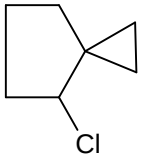
(C) 

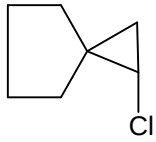
(D) 

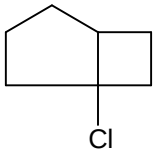
Space for Rough work

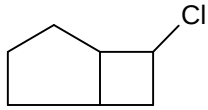
39. Major product obtained by the following reaction is



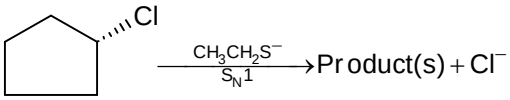
(A) 

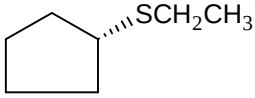
(B) 

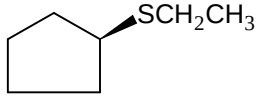
(C) 

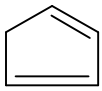
(D) 

40. What could be the product(s) for the following S_N1 reaction?



i) 

ii) 

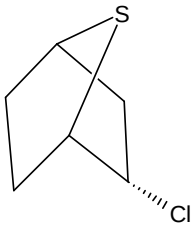
iii) 

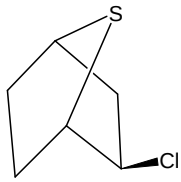
iv) 

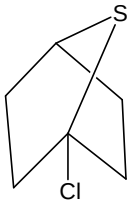
(A) i, ii
(B) i, iii
(C) iii, iv
(D) ii, iii

Space for Rough work

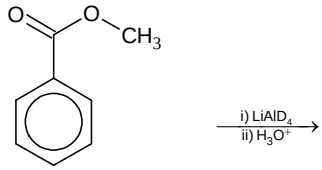
41. Identify the correct rate of hydrolysis

i) 

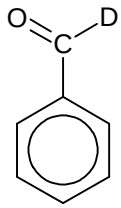
ii) 

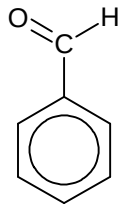
iii) 

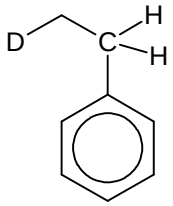
(A) i > ii > iii
(B) ii > iii > i
(C) i > iii > ii
(D) iii > i > ii

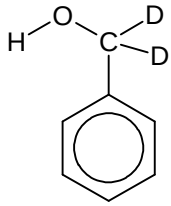
42. 

Choose the major product of the following reaction?

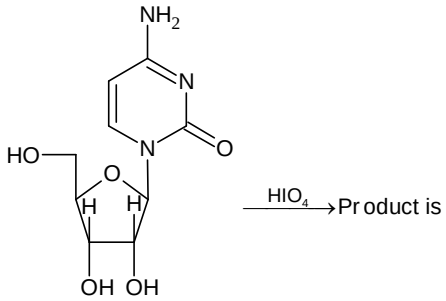
(A) 

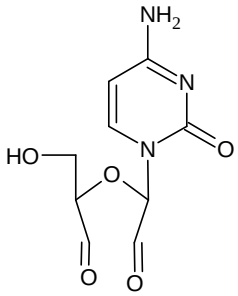
(B) 

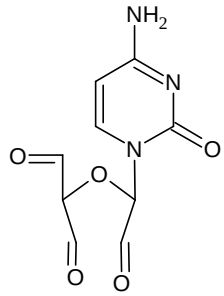
(C) 

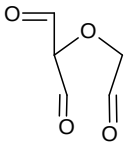
(D) 

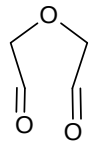
Space for Rough work

43. 

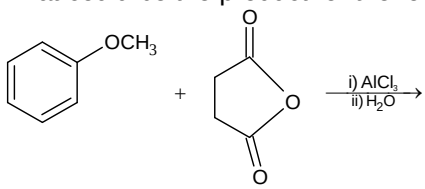
(A) 

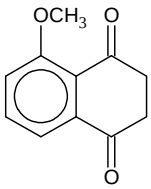
(B) 

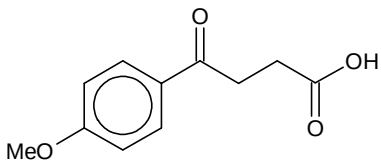
(C) 

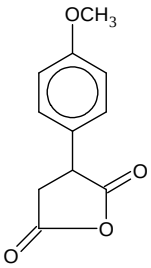
(D) 

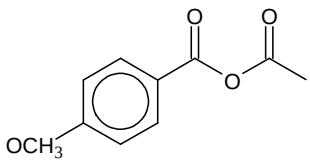
44. What could be the product for the following reaction?



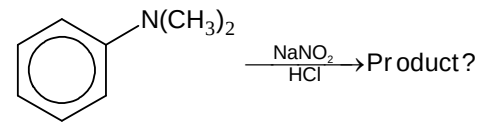
(A) 

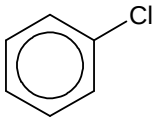
(B) 

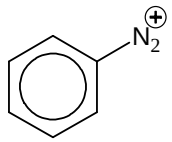
(C) 

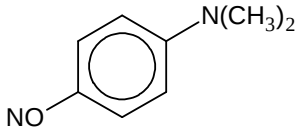
(D) 

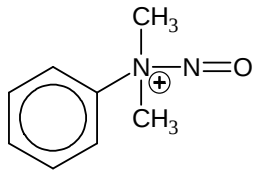
Space for Rough work

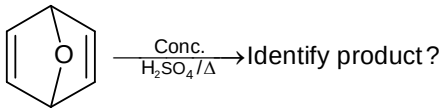
45. 

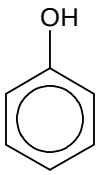
(A) 

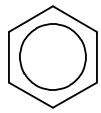
(B) 

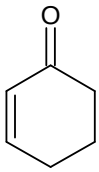
(C) 

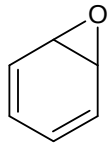
(D) 

46. 

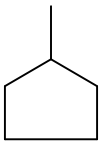
(A) 

(B) 

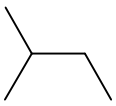
(C) 

(D) 

47. Consider the following reaction:



$\xrightarrow[h\nu]{Cl_2}$ Total number of mono chlorinated product = x



$\xrightarrow[h\nu]{Cl_2}$ Total number of mono chlorinated product = y

(Excluding stereoisomer)

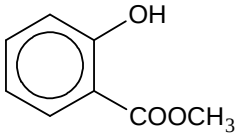
Identify value of x + y?

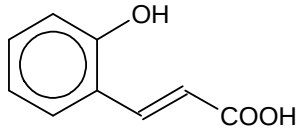
(A) 13 (B) 14

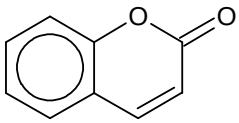
(C) 12 (D) 16

Space for Rough work

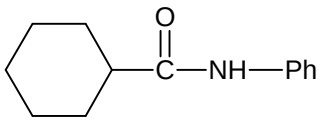
48. When o-hydroxybenzaldehyde is heated with ethanoic anhydride in the presence of sodium ethanoate, compound formed during the reaction is?

(A) 

(B) 

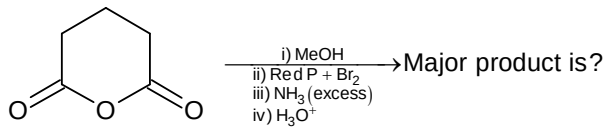
(C) 

(D) Both B and C

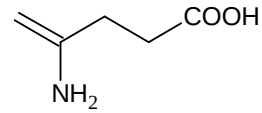
49. The IUPAC name of 

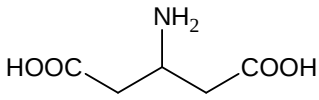
is

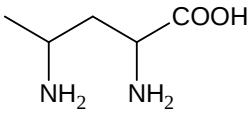
(A) N-cyclohexylbenzene
(B) N-Phenyl-N-cyclohexyl methanamide
(C) N-Phenylcyclohexanecarboxamide
(D) N-cyclohexyl-N-phenyl methyl amide

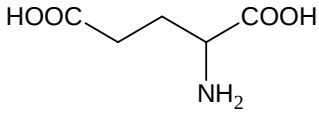
50. 

Major product is?

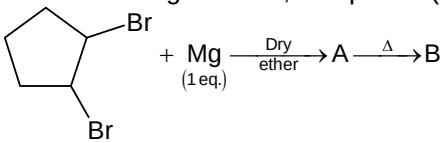
(A) 

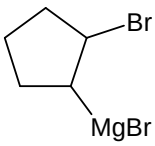
(B) 

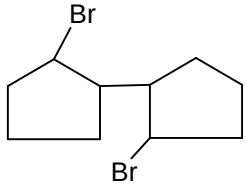
(C) 

(D) 

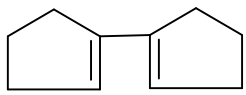
51. In the following reaction, compound (B) is



(A) 

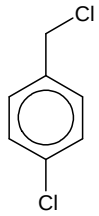
(B) 

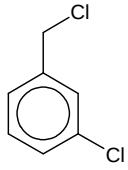
(C) 

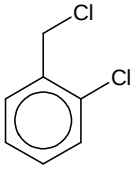
(D) 

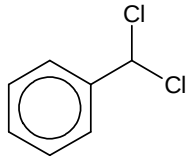
Space for Rough work

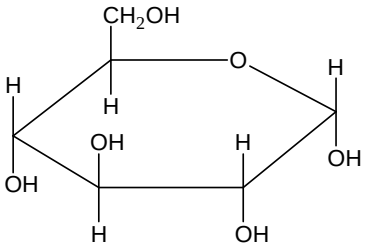
52. An aromatic compound (A), $C_7H_6Cl_2$ gives AgCl on boiling with alcoholic $AgNO_3$ soln. and yields C_7H_7OCl on treatment with NaOH. (A) on oxidation gives a monochlorobenzoic acid which affords only one monoderivative on nitration. The compound (A) is

(A) 

(B) 

(C) 

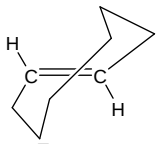
(D) 

53. 

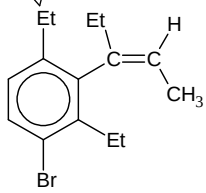
$\xrightarrow[H_2O]{Br_2}$ (A); number of chiral centre in compound (A) is :

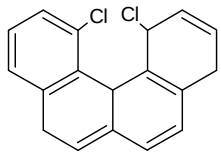
(A) 3
(B) 5
(C) 4
(D) 6

54. Which of the following is an optically inactive compound?

(A) 

(B) *Cis*-1,2-dimethylcyclohexane

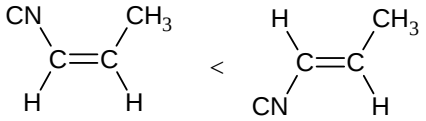
(C) 

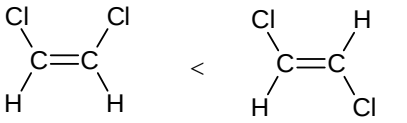
(D) 

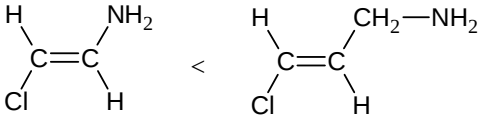
Space for Rough work

55. Which of the following options is incorrect?

(A) pK_{a_1} cis but-2-ene dioic acid > Trans but-2-ene dioic acid

(B) Dipole moment 

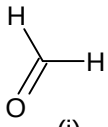
(C) Stability 

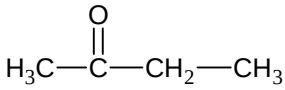
(D) Basicity 

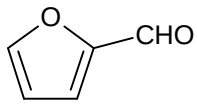
56. An optically active compound (A) has the molecular formula C_6H_{10}. The compound gives a ppt. when treated with $Ag(NH_3)_2OH$. On catalytic hydrogenation, A yields $B(C_6H_{14})$ which is only optically inactive. Identify the total number of αH in product formed by treatment of A with O_3/H_2O_2 then LAH and then $H^{\oplus}/\Delta$.

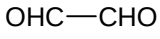
(A) 7 (B) 6
(C) 8 (D) 9

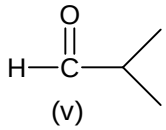
57. How many compounds will give Cannizzaro reaction?

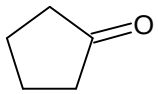

(i)


(ii)


(iii)


(iv)

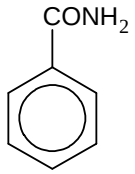

(v)

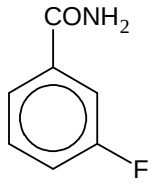

(vi)

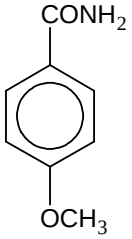
(A) 2 (B) 3
(C) 4 (D) 5

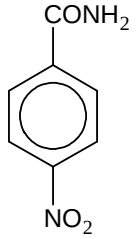
Space for Rough work

58. Which of the following can undergo Hofmann Bromamide reaction most easily?

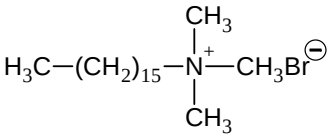
(A) 

(B) 

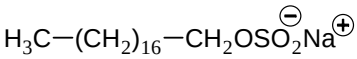
(C) 

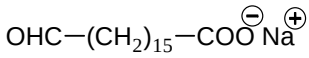
(D) 

59. Which of the following is not a surfactant?

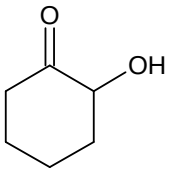
(A) 

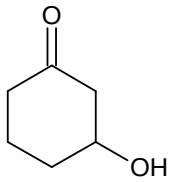
(B) $H_3C-(CH_2)_{14}-NH_2$

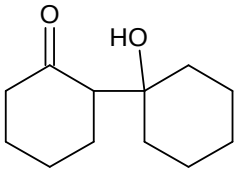
(C) 

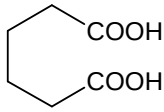
(D) 

60. When cyclohexanone is treated with Na_2CO_3 soln, we get

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A

(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

61. If tan α, tan β are the roots of the equation $x^2 + px + q = 0$, then the value of $\sin^2(\alpha + \beta) + p \sin(\alpha + \beta)\cos(\alpha + \beta) + q \cos^2(\alpha + \beta)$ is
(A) independent of p but dependent on q (B) independent of q but dependent on p
(C) independent of both p and q (D) dependent on both p and q

62. If the equation $x^3 - (1 + \cos\theta + \sin\theta)x^2 + (\cos\theta \sin\theta + \cos\theta + \sin\theta)x - \sin\theta \cos\theta = 0$ has roots α, β and γ then $\alpha^2 + \beta^2 + \gamma^2$ is
(A) 2 (B) $\frac{1}{2}$
(C) $\frac{3}{2}$ (D) none of these

63. If $x = \tan(22.5)^o$, then the value of $x^4 + 3x^3 - 3x^2 - 9x + 6$ is
(A) 0 (B) 3
(C) 2 (D) none of these

64. If x and y are non zero real numbers satisfying $xy(x^2 - y^2) = x^2 + y^2$, then the minimum value of $x^2 + y^2$ is
(A) 4 (B) $\frac{1}{4}$
(C) 3 (D) none of these

65. If the roots of $x^2 + ax + b = 0$ are $\sec^2\frac{\pi}{8}$ and $\text{cosec}^2\frac{\pi}{8}$, then which of the following is correct?
(A) a – b = 0 (B) a + b = 0
(C) 2a = b (D) a = 2b

Space for rough work

66. In a ΔABC, a semi circle is inscribed, whose diameter lies on the side c. If x is the length of the angle bisector through angle C, then the radius of the semicircle is (where s = semi perimeter and Δ = area of triangle)

(A) $\frac{\Delta}{x}$ (B) $\frac{abc}{2R^2(\sin A+\sin B)}$

(C) $x\sin\frac{C}{2}$ (D) $\frac{2\sqrt{s(s-a)(s-b)(s-c)}}{s}$

67. Let x be the set of all solutions to the equation $\cos x\cdot\sin\left(x+\frac{1}{x}\right)=0$. Number of real numbers contained by x in the interval $(0 < x < \pi)$, is

(A) 0 (B) 1
(C) 2 (D) more than 2

68. Let $f(x)=\frac{\text{cosec}^4x-2\text{cosec}^2x+1}{\text{cosec } x(\text{cosec } x-\sin x)+\frac{\sin x-\cos x}{\sin x}+\cot x}$. The sum of all the solutions of f(x) = 0 in $[0, 100\pi]$ is

(A) 2550π (B) 2500π
(C) 5000π (D) 5050π

69. In a ΔABC, $a = a_1 = 2$, $b = a_2$, $c = a_3$ such that $a_{p+1}=\frac{5^p}{3^{2-p}}a_p\left(2^{2-p}-\frac{4p-2}{5^p}a_p\right)$, where p = 1, 2 and r_1, r_2, r_3 are ex-radii, then

(A) $r_1 = r_2$ (B) $r_3 = 2r_1$
(C) $r_2 = 2r_1$ (D) $r_2 = 3r_1$

70. The value of $\tan\left(\sum_{r=1}^{\infty}\tan^{-1}\left(\frac{4}{4r^2+3}\right)\right)$ is equal to

(A) 1 (B) 2
(C) 3 (D) 4

71. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x. The maximum area enclosed by park is

(A) $\sqrt{\frac{x^3}{8}}$ (B) $\frac{1}{2}x^2$

(C) πx^2 (D) $\frac{3}{2}x^2$

Space for rough work

72. The equation of circle having the lines $x^2 + 2xy + 3x + 6y = 0$ as its normals and having size just sufficient to contain the circle $x^2 + y^2 - 4x - 3y = 0$ is $x^2 + y^2 + 6x - 3y - k = 0$ then k is
(A) 42 (B) 45
(C) $3\sqrt{5}$ (D) 15

73. The acute angle of a rhombus whose side is geometric mean of its diagonals is
(A) 15º (B) 20º
(C) 30º (D) 80º

74. The number of solutions of $\left(\log_{\sqrt{3}} \tan x\right)\left(\sqrt{\log_{\sqrt{3}} 3\sqrt{3} + \log_{\tan x} 3}\right) = -1$ in $[0, \pi]$ is
(A) 0 (B) 1
(C) 2 (D) 4

75. Vertex A of triangle ABC moves in such a way that tan B + tan C = constant then locus of orthocentre of ΔABC (BC is fixed) is a/an
(A) straight line (B) parabola
(C) circle (D) ellipse

76. If a hyperbola is confocal and coaxial with ellipse $\frac{x^2}{4} + y^2 = 1$ and intersect it at $\left(\sqrt{3}, \frac{1}{2}\right)$, then length of transverse axis of hyperbola is
(A) 4 (B) 2
(C) 3 (D) 7

77. The number of rational points on a circle with centre $\left(\sqrt{2}, -\sqrt{2}\right)$ and passing through (1, –1) are
(A) 1 (B) 2
(C) 4 (D) infinite

78. The values of λ for which the curve $\lambda(x^2 + y^2 + 2y + 1) = (x - 2y + 3)^2$ represents a hyperbola is
(A) $\lambda > 5$ (B) $0 < \lambda < 5$
(C) $\lambda < 0$ (D) $\lambda > 6$

79. If in a right angled triangle having integer sides a, b, $c \le 10$ the perimeter of the triangle is equal to the area of the triangle then area of triangle is
(A) 12 (B) 18
(C) 24 (D) 30

Space for rough work

80. The two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them is 60º. If the third side is 3 then the remaining fourth side is
(A) 2 (B) 3
(C) 5 (D) 8

81. Let A, B, C be three points on the ellipse $\frac{x^2}{4}+\frac{y^2}{1}=1$. If equation of AB is y = x then maximum area of ΔABC is
(A) 1 (B) $2\sqrt{2}$
(C) 2 (D) none of these

82. Let P(1, 1) is a point inside the circle $x^2 + y^2 + 2x + 2y - 8 = 0$. The chord AB is drawn passing through the point P. If $\frac{PA}{PB}=\frac{\sqrt{5}-2}{\sqrt{5}+2}$, then equation of chord AB is
(A) y = 2x + 1 (B) y = x
(C) y = –x (D) $y=\sqrt{2}x+1$

83. If AB is a double ordinate of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that ΔABC is equilateral, C being centre of hyperbola, then eccentricity e of hyperbola satisfies
(A) $e=\frac{2}{\sqrt{3}}$ (B) $e=\frac{\sqrt{3}}{2}$
(C) $1<e<\frac{2}{\sqrt{3}}$ (D) $e>\frac{2}{\sqrt{3}}$

84. Lines are drawn from a point P(–1, 3) to a circle $x^2 + y^2 - 2x + 4y - 8 = 0$ which meets the circle at two points A and B, then the minimum value of PA + PB is
(A) 6 (B) 8
(C) 10 (D) 12

85. A variable line ax + by + c = 0, where a, b, c are in A.P., is normal to a circle $(x-\alpha)^2 + (y-\beta)^2 = \gamma$, which is orthogonal to the circle $x^2 + y^2 - 4x - 4y - 1 = 0$ then the value of $\alpha + \beta + \gamma$ is
(A) 3 (B) 5
(C) 10 (D) 7

Space for rough work

86. Tangent is drawn to the ellipse $\frac{x^2}{27}+y^2=1$ at $\left(3\sqrt{3}\cos\theta, \sin\theta\right)$ (where $\theta \in \left(0, \frac{\pi}{2}\right)$), then the value of θ such that sum of intercepts on the axes made by this tangent is minimum, is

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{8}$ (D) $\frac{\pi}{4}$

87. The locus of orthocentre of the triangle formed by the focal chord of the parabola $y^2 = 4ax$ and the normals drawn at its extremities is

(A) $y^2 = a(x - 3a)$ (B) $y^2 = a(x + 3a)$

(C) $y^2 = a(x - 4a)$ (D) $y^2 = a(x + 4a)$

88. If the ellipse $\frac{x^2}{a^2-3}+\frac{y^2}{a+4}=1$ is inscribed in a square of side length $a\sqrt{2}$ then a is

(A) 4 (B) 2

(C) 1 (D) none of these

89. In the triangle ABC if sin C cos 15º cos A + sin 15º sin A = 1 then sin C + tan A tan B is equal to

(A) 1 (B) 2

(C) $\frac{3}{2}$ (D) $\frac{1}{2}$

90. From origin two perpendicular lines are drawn to intersect $\frac{x^2}{4}+\frac{y^2}{1}=1$ at A and B then $\frac{1}{(OA)^2}+\frac{1}{(OB)^2}$ is equal to

(A) 1 (B) $\frac{1}{4}$

(C) $\frac{5}{4}$ (D) $\frac{3}{4}$

Space for rough work